

Software Engineering & Project Management Documentation

Document 06: Change Management Plan & Change Control Governance

Document Control & Metadata

- **Project Name:** PharmaCart — Omni-Channel Retail Pharmacy Ordering & Inventory Platform
- **Document Identifier:** PC-CMP-DOC-006
- **Version:** 1.0 (Baseline Proposal - Pending Sign-Off)
- **Author / CCB Chair:** Ritesh Jadhav, Lead Project Manager & Systems Architect
- **Target Audience:** Project Steering Committee, Executive Leadership, Store General Managers, Engineering & QA Teams

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1. Executive Summary & Purpose

1.1 Purpose of the Change Management Plan

In a distributed retail healthcare software initiative spanning 14 physical pharmacies across Ahmedabad, project requirements cannot remain completely static. Market conditions, competitive forces, pharmacy council directives, and marketing initiatives will inevitably prompt stakeholders to request enhancements, architectural revisions, and operational deviations.

Uncontrolled requirement volatility—commonly referred to as **Scope Creep**—is the primary failure mode of software engineering projects. The purpose of this **Change Management Plan (CMP)** is to establish a rigorous, transparent, and defensible governance mechanism. It ensures that:

- Every requested change is captured, documented, and justified by business value.
- The multi-dimensional impact on **Scope, Schedule, Budget, System Quality**, and **Regulatory Compliance** is mathematically analyzed prior to execution.
- The baseline Core MVP (72 Story Points, 12 Weeks, ₹8.80 Lakh) remains protected against arbitrary expansion.
- Approved modifications are executed with zero disruption to the active critical path.

1.2 Boundary of Change Control

This protocol governs all modifications affecting:

1. **Requirements Baseline:** Functional and non-functional specifications defined in 01_SRS_Document.md.
2. **Architectural Baseline:** Data models, interfaces, and state machines established in 02_UML_Design_Package.md.
3. **Project Schedule & Budget:** Sprint allocations, milestones, and cash outlays established in 03_Project_Estimation_and_Plan.md.
4. **Quality Gates & Release Baselines:** Acceptance criteria and test coverage defined in 04_Test_Plan_and_QA.md.

2. Change Control Board (CCB) Charter

2.1 Governance Structure & Membership

The **Change Control Board (CCB)** is the formal steering entity authorized to evaluate, approve, defer, or reject all proposed modifications to project baselines.

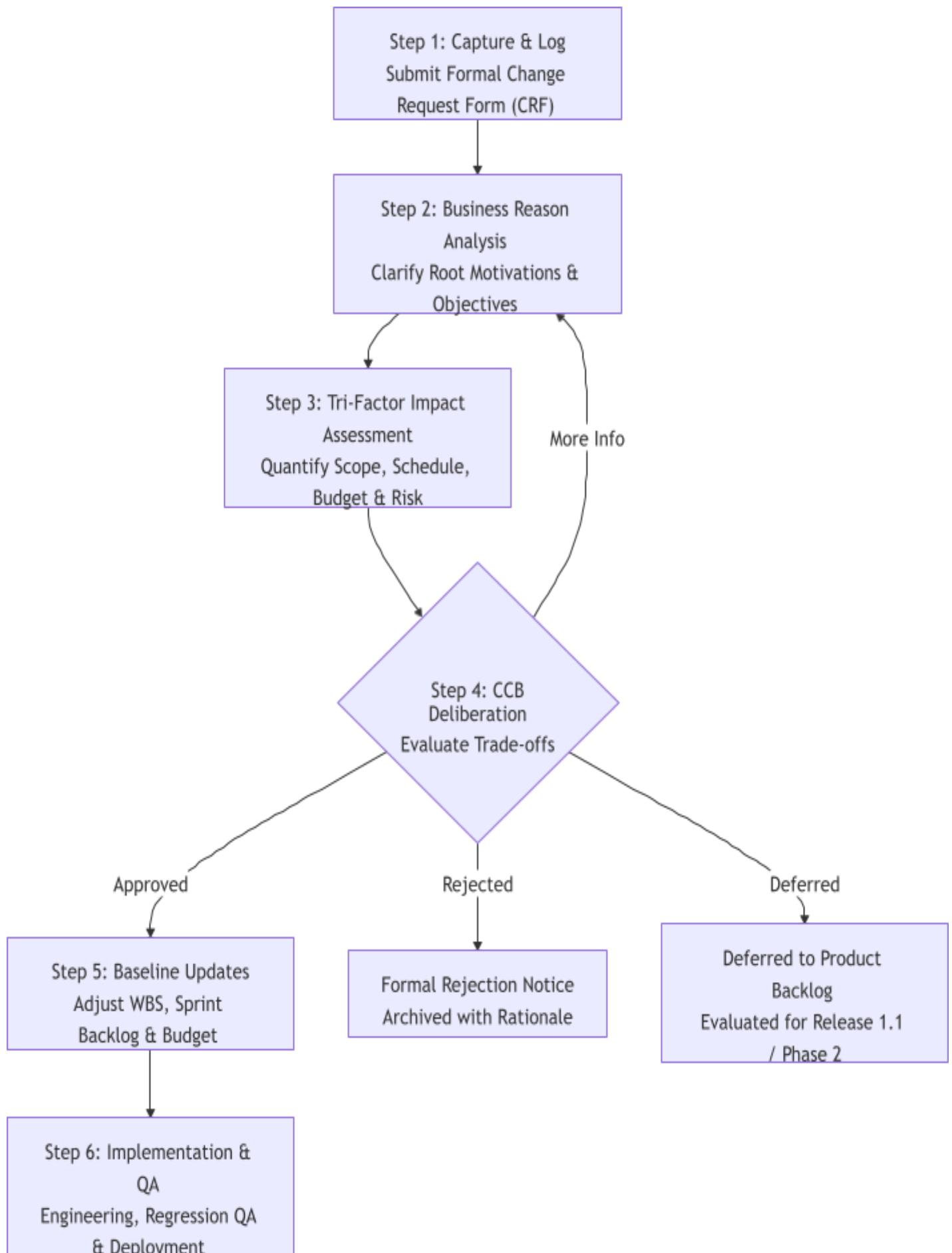
CCB Role	Designee / Title	Primary Responsibilities	Voting Authority
CCB Chair & Project Manager	Ritesh Jadhav	Convenes meetings, enforces CMP compliance, oversees impact analysis, and holds final execution authority.	1 Vote (Veto & Casting)
Chief Pharmacist & Compliance	Dr. A. K. Patel (VP Clinical)	Evaluates Schedule H/H1 legality, tele-pharmacy SLA impact, and GSP adherence.	1 Vote (Veto on Compliance)
Lead Software Architect	Technical Lead	Assesses architectural coupling, POS sync latency, scalability, and code maintainability.	1 Vote
Director of Store Operations	Retail Operations GM	Assesses counter staff friction, store hardware constraints, and Ahmedabad pilot readiness.	1 Vote
VP of Marketing & Sales	Commercial Lead	Advocates business ROI, customer acquisition drivers, and promotional campaign alignment.	1 Vote
Lead QA & Reliability Engineer	QA Test Lead	Quantifies regression testing effort, test suite expansion, and defect risk exposure.	Advisory (Non-Voting)

2.2 Quorum & Decision Rules

- **Quorum:** A valid CCB meeting requires the presence of at least 4 voting members, including the CCB Chair and Chief Pharmacist.
- **Approval Threshold:** Formal approval requires a majority vote (≥ 3 of 5 voting members) **and** zero regulatory vetoes from the Chief Pharmacist.
- **The Clinical Veto Rule:** If any change violates Gujarat State Pharmacy Council directives, Drugs and Cosmetics Act regulations, or patient safety mandates, the Chief Pharmacist holds absolute unilateral veto authority.
- **The Fiscal & Schedule Ceiling Rule:** If any change threatens to breach the hard project ceiling of **₹14.0 Lakh** or the **5-Month calendar deadline**, the CCB Chair must reject or defer the request to post-MVP operations unless additional capital and time are formally injected by the Executive Sponsor.

3. The 6-Step Formal Change Control Lifecycle

The PharmaCart project adheres to a disciplined 6-stage lifecycle for all change proposals.



Step 1: Capture & Submission

- Any project stakeholder (Marketing, Store Managers, Clinical Staff, Engineering) may initiate a change by submitting a standardized **Change Request Form (CRF)** .
- Verbal, chat, or informal requests are strictly prohibited from engineering backlogs.
- Within 24 business hours of receipt, the CCB Chair assigns a tracking ID (CR-XX) and records it in the **Master Change Request Log (CRL)** .

Step 2: Understand the Business Reason

- The CCB reviews the fundamental business drivers behind the request.
- Requests are categorized into one of five operational drivers:
 - Regulatory / Legal Compliance* (Mandatory).
 - Defect Mitigation / Operational Stability* (Urgent).
 - Competitive Advantage / Revenue Growth* (Commercial).
 - Customer Experience Enhancement* (Usability).
 - Internal Operational Efficiency* (Cost Reduction).
- If the underlying motivation lacks clear business metrics (e.g., projected customer retention, regulatory penalty avoidance), the request is returned for clarification.

Step 3: Tri-Factor Impact Assessment

Before the CCB convenes, a joint technical-business task force performs an exhaustive multi-dimensional impact assessment:

- Scope Impact:** Precise Delta in Story Points (+/- SP), affected WBS elements, and modified architectural modules.
- Schedule Impact:** Additional working days required, impact on Early Finish (EF) / Late Finish (LF) of critical path tasks, and projected sprint shift.
- Cost Impact:** Development labor outlay (Hours * ₹1,500/hr or Sprint burn), infrastructure overhead, and third-party API licensing costs.
- Quality & Risk Impact:** Additional regression test cases needed, test environment dependencies, and risk exposure delta (\$P \times I\$).

Step 4: CCB Deliberation & Disposition

The CCB meets on a scheduled bi-weekly cadence (or within 48 hours for Critical severity requests) to review the impact assessment. The board renders one of four binding verdicts:

- Approved for Current Release:** The change is integrated into the active sprint plan, with compensatory scope de-prioritization if schedule slippage threatens milestones.
- Deferred to Next Release (Phase 2):** The change is validated as valuable but scheduled for post-MVP execution (Release 1.1) to safeguard the 12-week Core MVP launch date.
- Rejected:** The change contradicts architectural principles, incurs disproportionate risk, or provides insufficient business return on investment.
- Request More Information (RMI):** The technical or commercial analysis requires further data before a determination can be made.

Step 5: Update Baselines & Plans

Upon approval, the CCB Chair and Technical Leads execute formal configuration updates:

- SRS Update:** New Functional Requirements (FRs) added; Requirement Traceability Matrix (RTM) updated.
- Sprint Backlog Revision:** Approved user stories placed in designated sprint backlogs.

- **WBS & Gantt Revision:** Activity durations and precedence links re-calculated; baseline version incremented (e.g., Baseline v1.0 to Baseline v1.1).
- **Budget Tracking:** Allocated contingency funds deducted from the ₹5.20 Lakh reserve.

Step 6: Implementation, QA Verification & Formal Closure

- Engineering builds the feature under an isolated Git feature branch (`feat/cr-xx-description`).
- The QA team executes targeted unit tests, integration suites, and complete regression tests per `04_Test_Plan_and_QA.md`.
- The CCB Chair verifies that acceptance criteria and NFR SLAs are verified in staging.
- The change is formally closed, deployed, and marked as `VERIFIED & CLOSED` in the CRL.

4. Change Request Template & Submission Artifacts

To maintain rigorous configuration control, all stakeholders must utilize the standardized PharmaCart Change Request Form.

Field Identifier	Field Name	Description & Mandatory Input Guidelines
CR-ID	Change Request Identifier	Unique sequential tracking identifier (e.g., CR-01 , CR-02).
DATE-SUBMITTED	Submission Date	Calendar date of formal logging (YYYY-MM-DD).
INITIATOR	Requesting Stakeholder	Name, departmental title, and contact details of the requester.
TITLE	Change Summary	Concise, descriptive title of the proposed capability.
DESCRIPTION	Detailed Specification	Exact description of current baseline behavior vs. proposed modification.
BUSINESS JUSTIFICATION	Business Driver & Value	Concrete revenue, compliance, customer retention, or operational rationale.
SEVERITY / URGENCY	Priority Classification	<i>Critical</i> (Breaks law/production), <i>High</i> (Major value), <i>Medium</i> (Standard), <i>Low</i> (Cosmetic).
TARGET PHASE	Desired Delivery Window	Core MVP (Release 1 .0) vs. Phase 2 (Release 1 .1).

5. In-Depth Case Study: Marketing Request (CR-01)

5.1 The Scenario: Marketing's High-Pressure Demand

At the conclusion of Sprint 1, the Vice President of Marketing submitted a high-priority change proposal:

- **Feature Request:** Implement a **Chronic Medicine Auto-Refill Subscription Engine** and **Tiered Promotional Discount / Coupon System** immediately in Release 1.0.
- **Marketing Justification:** Competitors (Tata 1 mg, Apollo 24/7) offer auto-refill subscriptions for diabetic and hypertensive patients. Marketing projected that launching without recurring subscriptions would sacrifice 35% of high-LTV (Lifetime Value) chronic care customer acquisition during the initial launch quarter.

- **Initial Pressure:** Marketing demanded that engineering absorb this request into the 12-week MVP timeline without shifting the launch date.

5.2 Step 1 & 2: Change Proposal Capture & Business Motivation

Attribute	Specification Details
Change ID	CR-01
Initiator	VP of Marketing & Commercial Strategy
Date Logged	2026-10-19 (End of Sprint 1)
Title	Chronic Medicine Recurring Auto-Refill Subscriptions & Dynamic Coupon Engine
Business Objective	Maximize customer retention for high-margin chronic ailments (Cardiology, Diabetology) across Ahmedabad.
Target Audience	~4,200 recurring chronic medication buyers currently visiting the 14 physical stores.

5.3 Step 3: Rigorous Tri-Factor Impact Assessment

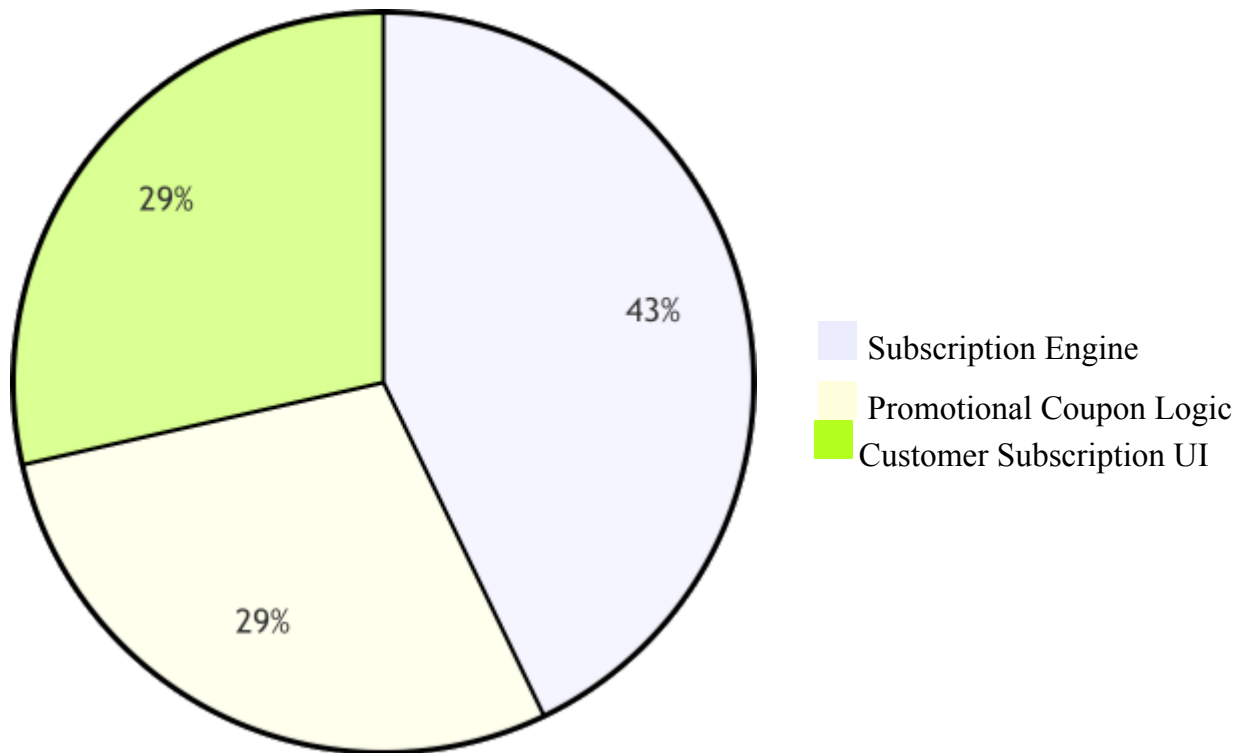
The CCB technical task force, chaired by **Ritesh Jadhav**, conducted a 48-hour technical and operational evaluation.

A. Scope Impact (Work Packages & Complexity Analysis)

Implementing CR-01 requires three substantial new functional modules:

1. **Recurring Subscription Scheduler (Backend):** Cron-like scheduler to calculate re-order intervals (30, 60, 90 days), trigger pharmacist review 5 days before exhaustion, and handle tokenized recurring billing.
2. **Promotional Discount & Cart Rule Engine (FinTech):** Minimum order value validations, category restrictions (discounting Schedule H drugs is strictly regulated), usage limits per customer, and coupon abuse protection.
3. **Customer Subscription Management UI (Frontend):** Mobile web portal views to pause, modify delivery dates, alter dosages, and update stored payment methods.

CR-01 Story Point Breakdown (28 SP Total)



- **Total Additional Effort: 28 Story Points.**
- **Relative Scope Expansion:**

Baseline Core MVP Scope: 72 Story Points
 CR-01 Scope Addition: 28 Story Points
 Total Proposed Scope: 100 Story Points (+38.8% Increase)

B. Schedule Impact (Critical Path & Velocity Analysis)

PharmaCart's empirically proven development velocity is **14 Story Points per 2-week Sprint** (7 Story Points/week).

- **Additional Development Time:**

Additional Sprints = 28 Story Points / 14 Story Points per Sprint = 2.0 Full Sprints (4 Calendar Weeks)

- **Critical Path Collision:**
 - Baseline Core MVP completion: Week 12 (Month 3).
 - If absorbed into Release 1.0: Launch date slips from **Week 12 to Week 16** (Month 4).
 - Schedule buffer drops from 9.3 weeks down to only **5.3 weeks**.

C. Budget & Cost Impact (Labor & Reserves)

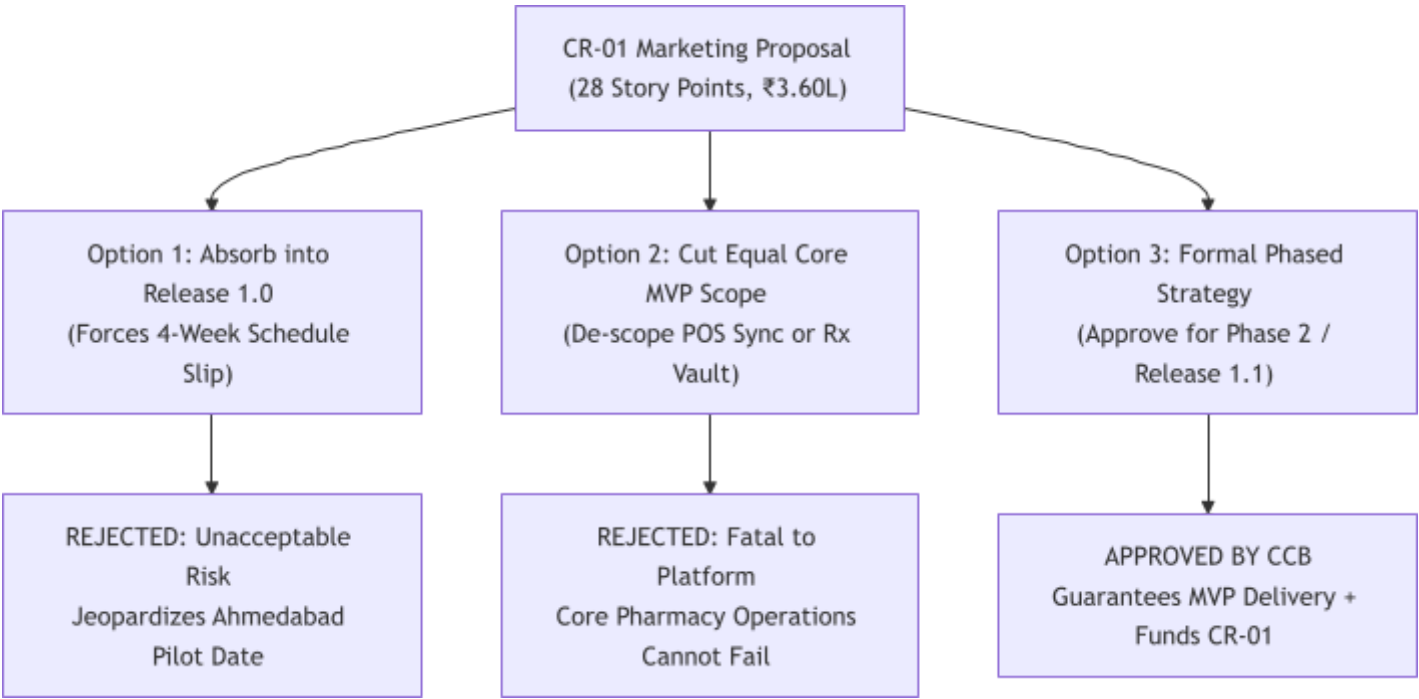
- **Development Labor Burn:** 2 additional sprints = 1 month of dedicated team burn = ₹2,40,000 / month * 1.0 Month = ₹2,40,000.
- **Third-Party SaaS Overhead:** Recurring payment mandate tokenization (Razorpay Subscriptions API) and high-volume transactional WhatsApp alert packs = ₹40,000.
- **Targeted QA & Security Auditing:** Penetration testing for coupon tampering and recurring billing logic = ₹80,000.
- **Total Financial Cost of CR-01 : ₹3,60,000.**

D. Regulatory & Operational Risk Impact

- **Chief Pharmacist Assessment:** Under the Drugs and Cosmetics Rules, Schedule H prescription refills **cannot be automated unconditionally**. A licensed pharmacist must verify that the doctor's original prescription remains valid, that maximum refill counts have not been exceeded, and that the patient has not reported adverse drug reactions. Automating dispatch without human clinical sign-off would violate state pharmacy regulations.

5.4 Step 4: CCB Deliberation, Decision & Trade-Off Modeling

The CCB convened on 2026-10-21 to evaluate the three possible paths:



Detailed Trade-Off Matrix Evaluated by CCB:

Evaluation Dimension	Option 1 : Cram into Release 1 .0	Option 2: Swap Scope (1 - for-1 Cut)	Option 3: Phased Release 1 .1 (Selected)
Release 1 .0 Launch Date	Slipped to Week 1 6 (4-Week Delay)	Week 1 2 (Preserved)	Week 1 2 (1 00% Preserved)
Core Operational Integrity	Strained; QA condensed by 50%	Catastrophic; requires cutting POS sync	Flawless; 1 00% Core QA executed
Budget Position	Consumes ₹3.6L; reserves drop to ₹1 .6L	Neutral	Funded from ₹5.20L reserve; leaves ₹1 .60L
Pharmacy Council Compliance	High legal liability (rushed refill logic)	Acceptable	1 00% Compliant; includes human gate
Marketing Satisfaction	Delayed total platform launch	Core customers alienated	Gets fully robust engine in Month 4

The CCB Binding Ruling:

1. **Unanimous Disposition: APPROVED FOR PHASE 2 (RELEASE 1.1).**

2. Defensible Rationale:

- Release 1.0 (Core MVP) will launch on schedule at **Week 12** containing all 72 baseline story points (Prescription Desk, 14-Store POS Sync, 2-Unit Safety Buffer, In-Store Pickup).
- Sprints 7 and 8 (Weeks 13–16) are formally commissioned as **Phase 2 (Release 1.1 : Loyalty & Retention Expansion)**.
- Development of CR-01 will commence immediately upon successful completion of the 3-store Ahmedabad pilot in Week 11.
- The financial outlay of **₹3,60,000** is formally allocated from the **₹5,20,000 unallocated project cash reserve**, leaving an uncommitted liquid buffer of **₹1,60,000** for emergency hardware contingencies.
- The feature architecture is updated to mandate an explicit **Pharmacist Clinical Re-authorization Checkbox** prior to every recurring monthly dispense, neutralizing regulatory liability.

5.5 Step 5: Baselines & Configuration Updates

Following CCB approval, the following formal document modifications were recorded:

- **WBS Update:** Created Work Package WBS 6.0: Chronic Care & Commercial Engine with sub-tasks 6.1 (Subscription Engine), 6.2 (Coupon Rule Matrix), and 6.3 (Customer Portal Controls).
- **RTM Update:** Linked CR-01 to Functional Requirements FR-13: Chronic Refill Management and FR-14: Dynamic Promotional Engine.
- **Budget Ledger Update:**

Total Available Capital:	₹14,00,000
Baseline Core MVP Expenditure:	- ₹8,80,000
Unallocated Capital Prior to CR-01:	₹5,20,000
CR-01 Phase 2 Allocation:	- ₹3,60,000
Remaining Liquidity Contingency:	₹1,60,000

5.6 Step 6: Implementation, QA & Deployment Plan

The phased engineering schedule for CR-01 seamlessly bridges Release 1.0 stabilization and Release 1.1 go-live:

Milestone / Deliverable	Target Timeline	Verification Criteria & QA Gate	Sign-Off Authority
Architecture Specification	Week 1 1 (During QA)	Architecture doc update; PCI-DSS token storage design review	Lead Architect
Sprint 7 Engineering	Weeks 1 3–1 4	Scheduler daemon unit tests (>90% branch coverage)	Tech Lead
Sprint 8 Engineering	Weeks 1 5–1 6	Razorpay webhook reconciliation, coupon abuse penetration tests	Lead QA Engineer
Clinical Acceptance Gate	Week 1 6 (End)	Council verification gate signed off by Chief Pharmacist	Dr. A. K. Patel
Production Rollout (v1 .1)	Week 1 7 (Month 4)	Zero downtime deployment across all 1 4 Ahmedabad store portals	Ritesh Jadhav

6. Master Change Request Log (CRL)

The following log tracks all formal change requests submitted throughout the PharmaCart development lifecycle.

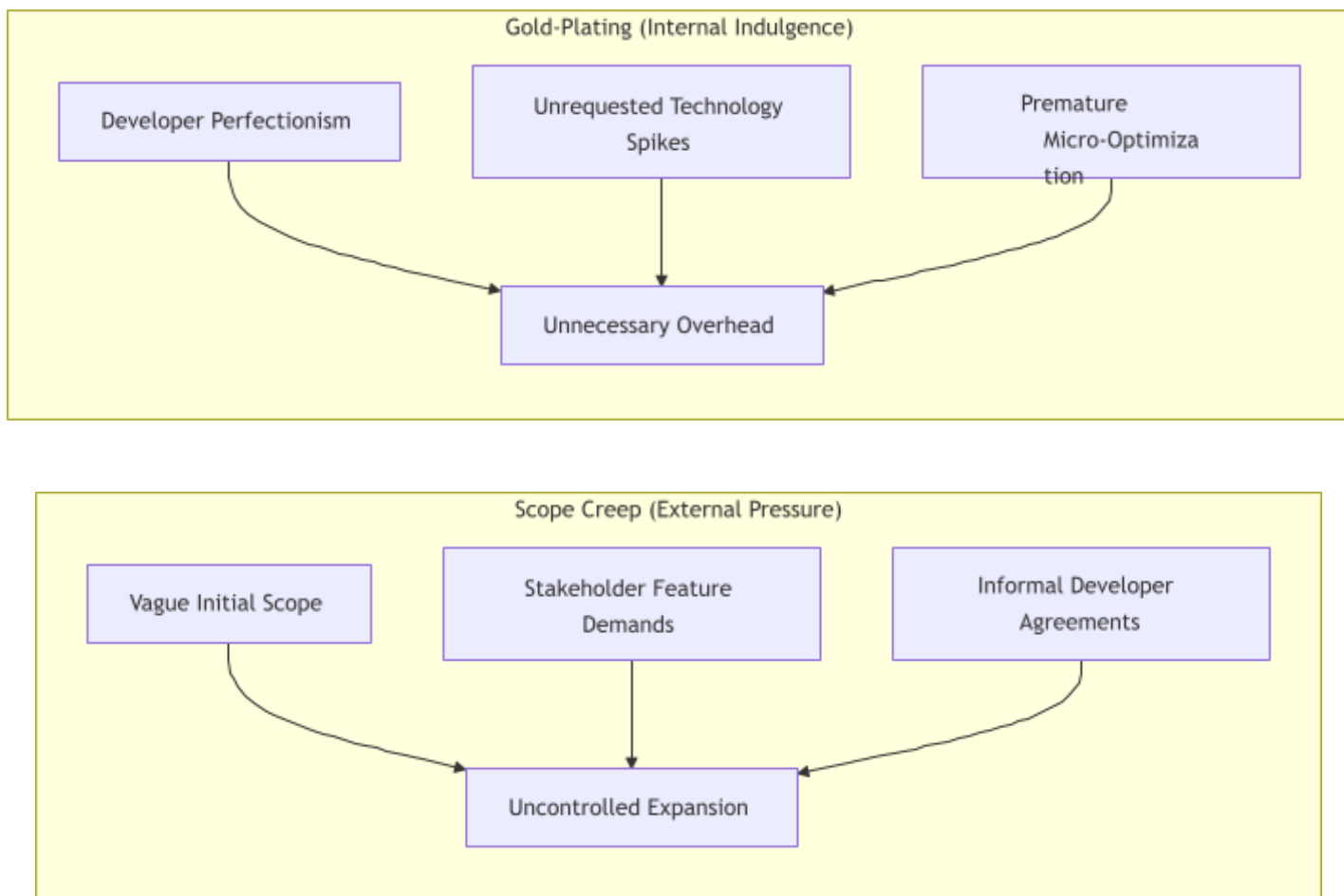
CR-ID	Date Logged	Request Title & Initiator	Scope Delta	Cost Impact	Schedule Impact	Status / Disposition	CCB Decision Date
CR-01	2026-10-19	Chronic Medicine Refill Subscriptions & Dynamic Coupon Engine <i>(VP Marketing)</i>	+28 SP	+ ₹3,60,000	+4 Weeks (Post-MVP)	Approved for Phase 2 (Release 1.1)	2026-10-21
CR-02	2026-11-04	Ayushman Bharat Digital Mission (ABDM / ABHA ID) Integration <i>(Compliance Dept)</i>	+14 SP	+ ₹1,80,000	+2 Weeks	Under Technical Evaluation	Pending Sandbox API Docs
CR-03	2026-11-12	Real-Time Delivery Rider GPS Tracking & Route Optimization <i>(Logistics Partner)</i>	+35 SP	+ ₹4,50,000	+5 Weeks	REJECTED (Out of Scope)	2026-11-14
CR-04	2026-11-20	Regional Language UI (Gujarati & Hindi Multilingual Localization) <i>(Ahmedabad Store Mgrs)</i>	+10 SP	+ ₹1,20,000	+1.5 Weeks	Deferred to Q2 FY27 Backlog	2026-11-22

Detailed Disposition Rationale for Logged Requests:

- 1. **CR-02 (ABDM / ABHA Health ID):** Evaluating whether integrating the Indian government's digital health identity system should be deployed during Phase 2. Decision deferred pending National Health Authority (NHA) milestone 2 sandbox stability.
- 2. **CR-03 (Delivery Rider GPS Tracking):** Unanimously **REJECTED**. PharmaCart's foundational value proposition is localized store pickup and third-party logistics dispatch. Building an in-house Uber-style rider fleet routing engine is classic gold-plating that directly violates our retail pharmacy mission.
- 3. **CR-04 (Multilingual Localization):** Validated as beneficial for retail counter customers in Gujarat, but existing mobile web users predominantly utilize English medicine names. Deferred to FY27 Q2 enhancement cycle.

7. Scope Creep & Gold-Plating Defense Strategy

7.1 Scope Creep vs. Gold-Plating: Definitions & Root Causes



- **Scope Creep (External):** The uncontrolled, unapproved addition of features by stakeholders without corresponding adjustments to budget, time, or resources.
- **Gold-Plating (Internal):** The unauthorized practice of engineers adding extra complexity, unrequested utility libraries, or over-engineered abstractions beyond what the customer specified, under the guise of "doing a thorough job."

7.2 The 5 Core Defense Policies for PharmaCart

To ensure zero scope creep and zero gold-plating during the 12-week construction window, the following governance policies are strictly enforced:

Policy 1 : Zero Informal Channels (The "No Hallway Agreement" Rule)

- Engineers are strictly forbidden from altering API contracts, database schemas, or UI elements based on verbal conversations, WhatsApp messages, or store manager requests.
- Any engineer who commits code addressing an unlogged requirement will have their pull request immediately rejected by the Tech Lead.

Policy 2: The "Trade-In" Requirement for Scope Additions

- If business leadership insists that an emergent feature must launch in Release 1.0, an **equivalent volume of story points must be explicitly removed** from the active sprint backlog:

Scope In = Scope Out (Delta SP = 0)

- For example, if a new 8-point store reporting widget is approved for Release 1.0, an existing 8-point feature (such as customer order history export or loyalty points display) must be decommissioned from Release 1.0.

Policy 3: Strict Definition of Done (DoD) Enforcement

Every user story in the PharmaCart backlog has explicit, legally binding acceptance criteria.

- A story is considered "Done" when its acceptance criteria pass—nothing more, nothing less.
- Pull requests that add unrequested features or unneeded third-party libraries will fail the Architecture Quality Gate.

Policy 4: The 15% Architectural Simplicity Rule (YAGNI & KISS)

- Engineers must adhere strictly to **YAGNI (You Aren' t Gonna Need It)** .
- Building complex generic plugin architectures, multi-tenant databases (when only 14 retail stores exist in a single tenant), or Kafka streaming clusters when a lightweight Redis pub/sub queue meets NFR SLAs is treated as an engineering defect.

Policy 5: Agile Backlog Pruning at Sprint Boundaries

- At the beginning of each 2-week sprint planning meeting, the Project Manager and CCB Chair reviews all logged user stories against the project' s primary KPI: **Launching 14 connected stores with reliable 30-second POS sync by Week 12.** Any task not contributing to this goal is immediately pruned.

8. Configuration Management & Traceability Integration

8.1 Baseline Management & Version Control Scheme

All project deliverables are maintained under formal semantic version control to ensure complete traceability.

Deliverable Type	Primary Artifact File	Baseline Version (Approved)	Next Revision Trigger
Requirements	01_SRS_Document.md	v1 .0	CCB-approved functional modifications
System Architecture	02_UML_Design_Package.md	v1 .0	Data schema or interface modifications
Project Plan & CPM	03_Project_Estimation_and_Plan.md	v1 .0	Sprint velocity recalibration or scope changes
QA Test Plan	04_Test_Plan_and_QA.md	v1 .0	Test case expansion for new releases
Risk Register	05_Risk_Register_and_RMMM.md	v1 .0	Monthly risk re-scoring or incident emergence
Change Control	06_Change_Management.md	v1 .0	CCB charter or policy updates

8.2 End-to-End Change Traceability Matrix

When a change request transitions to APPROVED , it is assigned end-to-end traceability across the entire engineering artifact suite:

```
CR-01 (Marketing Request)
|
├─> SRS Requirement:      FR-13 (Chronic Subscriptions) & FR-14 (Coupons)
├─> Architecture Module:  SubscriptionSchedulerDaemon & CouponValidatorService
├─> Test Plan Test Cases: TC-SUB-01 through TC-SUB-12 (Payment Tokenization & Cadence)
├─> Risk Register Entry:  RSK-05 (Automated Schedule H Refill Compliance Hazard)
└─> Project Estimation:   Phase 2 (Sprints 7 & 8, 28 Story Points, ₹3.60L Budget)
```

9. Formal CCB Sign-Off & Governance Endorsement

The following voting members of the **PharmaCart Change Control Board** are designated to review, ratify, and endorse this Change Management Plan. Formal baseline endorsement remains pending final convening and sign-off.

Role / Title	Board Member Name	Signature & Ratification Date	Approval Status
CCB Chair & Lead Project Manager	Ritesh Jadhav	Pending	Pending
Chief Pharmacist & Compliance VP	Dr. A. K. Patel	Pending	Pending
Lead Software Architect	Technical Lead	Pending	Pending
Director of Store Operations	Retail Operations GM	Pending	Pending
VP of Commercial Strategy & Marketing	Commercial Lead	Pending	Pending

End of Document 06: Change Management Plan & Change Control Governance